

Optic Nerve Head Anatomy and Vascular Risk Factors in Patients With Optic Disc Drusen Associated Anterior Ischemic Optic Neuropathy



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• **PURPOSE:** Optic disc drusen (ODD) is an anatomical risk factor for nonarteritic anterior ischemic optic neuropathy (NA-AION). This study aimed to investigate the anatomical and vascular risk factors of patients with ODD-AION (ODD-associated NA-AION) and compare them with similar data from patients with nODD-AION (NA-AION without ODD).

• **DESIGN:** : Case-control study.

• **METHODS:** Thirty-four ODD-AION and 34 nODD-AION patients who had all been systematically optical coherence tomography scanned using a standardized ODD scanning protocol were retrospectively analyzed and compared regarding demographics, vascular risk factors, clinical characteristics, and specific optic nerve head anatomical characteristics.

• **RESULTS:** In patients with ODD-AION, the ODD were predominantly deeply located (82%) but with no significant difference in size (52% large, 48% small). When compared with nODD-AION patients, ODD-AION patients were significantly younger at the time of diagnosis ($P = .012$) and had fewer vascular risk factors ($P = .015$). The ODD-AION patients had significantly more peripapillary hyperreflective ovoid mass-like structures (PHOMS) ($P < .001$) and prelaminar hyperreflective lines ($P < .001$) as well as smaller Bruch's membrane opening diameters ($P = .017$) compared with nODD-AION patients. No significant differences were found between ODD-AION and nODD-AION patients regarding visual acuity, refraction, lamina cribrosa position, ganglion cell layer volume, or retinal nerve fiber layer thickness.

• **CONCLUSION:** In ODD-AION, location of the ODD within the optic nerve head is important, while the size of the ODD is not. The ODD-AION and nODD-AION patients presented with distinctly different vascular risk factors and anatomical

characteristics, establishing ODD and potentially also PHOMS as independent risk factors for developing NA-AION. (Am J Ophthalmol 2022;242: 156–164. © 2022 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>))

INTRODUCTION

Anterior ischemic optic neuropathy (AION) is ischemic damage to the anterior portion of the optic nerve.¹ Nonarteritic AION (NA-AION) refers to small-vessel disease in which hypoperfusion of the short posterior ciliary arteries leads to ischemia, causing acute, painless, and monocular vision loss. Nonarteritic AION has been linked to several vascular risk factors such as hypertension, diabetes, and smoking. Furthermore, a crowded so-called “disk-at-risk” is an anatomical risk factor making the prelaminar optic nerve head (ONH) susceptible to local compartment syndrome, which is the presumed pathophysiology of NA-AION.^{1–3}

Optic disc drusen (ODD) are acellular deposits of calcium and mucopolysaccharides that can be found in the prelaminar ONH, with a prevalence of approximately 2% in the general population.⁴ Enhanced depth imaging optical coherence tomography (EDI-OCT) of the ONH has high sensitivity for detection and is now considered the gold standard for visualization of ODD.⁵ To date, the exact etiology remains unclear but ODD are thought to exist due to an axonopathy in which mitochondria are extruded into the prelaminar extracellular space,⁶ and/or due to a small Bruch's membrane opening (BMO) where axoplasmic flow is blocked via localized compression.⁴ The ODD are asymptomatic in most cases, yet peripheral visual field defects can be seen on examination in up to 87%.⁷ Nonarteritic AION is a well-known complication of ODD and it has recently been found that > 50% of NA-AION patients aged < 50 years have underlying ODD.^{2,8}

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Recognizing ODD as an independent anatomical risk factor for AION, the term ODD-AION has been introduced.⁸ Likewise, NA-AION without ODD can be termed non-ODD-AION (nODD-AION). This study aimed to examine and compare the ONH anatomy, using OCT, and vascular risk factors in patients with ODD-AION and nODD-AION. It hypothesized to find phenotypical differences besides the presence of ODD, potentially playing a role in the pathophysiology of NA-AION in the two groups.

MATERIALS AND METHODS

• **STUDY POPULATION:** A previous study compared 13 patients with ODD-AION with 14 patients with nODD-AION regarding vascular but not anatomical risk factors.³ These patients were diagnosed at Rigshospitalet, Glostrup, Denmark, in the period 2008-2017, termed inclusion period A. Briefly, eligible patients in inclusion period A were first identified using diagnostic coding. All patients with ODD-AION and the 20 youngest adult patients with nODD-AION were subsequently invited in an attempt to find age-matched patients. A study examination was then performed on the ODD-AION and nODD-AION patients from November 2017 to May 2018. During the first 2 months of data acquisition (November-December 2017) ODD-AION and nODD-AION patients who were seen in the Neuro-Ophthalmology clinics at Rigshospitalet were also enrolled. All patients from inclusion period A provided written informed consent.

In order to increase the number of participants in this retrospective study, the total inclusion period was expanded to also include 2017-2021, termed inclusion period B. The only demographic difference between the patients from inclusion period A and period B was that the latter also included patients aged < 18 years. The reason for doing this was to get as wide a variety of patient ages as possible, allowing focus on potential anatomical biomarkers in the two patient groups. This study was not interested in commenting on the prevalence of AION or ODD.

In inclusion period B, patients with an ICD10 diagnosis code for AION (DH470C) seen at the Department of Ophthalmology, Rigshospitalet, were systematically identified via social security numbers. Their OCTs were reviewed using Heidelberg Eye Explorer, and patients were categorized as ODD-AION or nODD-AION based on whether they had ODD or not. The ODD-AION and nODD-AION diagnoses were then cross-checked with the patient's medical records, in which information regarding other diagnoses, comorbidities, age, intraocular pressure (IOP), and visual acuity were also accessed. Access to patient files from inclusion period B was granted by the Center for Regional Development in the Capital Region of Denmark (protocol number R-21047751).

The inclusion and exclusion criteria applied both to patients in inclusion periods A and B, unless otherwise stated. The following inclusion criteria were used to ensure that NA-AION was diagnosed in at least one eye: 1) unilateral monocular vision loss, 2) unilateral optic disc edema during the acute state, 3) disc-related visual field loss, 4) dense ONH EDI-OCT scan cf. Optic Disc Drusen Studies (ODDS) Consortium scanning protocol (see below) performed at the study visit (patients from period A) or within 14 days of diagnosis (patients from period B), and 5) aged > 18 years (patients from period A only). Exclusion criteria were: 1) duplicates, 2) glaucoma, 3) ipsilateral ONH disease, 4) compressive lesion of the afferent visual pathways, 5) giant cell arteritis, 6) incorrect diagnosis, and 7) IOP > 21 mmHg at study examination.

The study was approved by the Scientific Ethics Committee of the Capital Region of Denmark (protocol number H-17021284, inclusion period A) and the Danish Data Protection Agency (protocol number P-2021-519, inclusion periods A and B). The study was performed in accordance with the Helsinki Declaration.

• **DATA ACQUISITION:** The EDI-OCT (Spectralis HRA + OCT; Heidelberg Engineering) was performed in accordance with the ODDS Consortium guidelines.⁵ Briefly, a dense ONH scan was obtained with a pattern size of 15° X 10° using 97 B-scans for identification and classification of ODD. A 20° 6-line radial scan was used to measure BMO diameter and volume of peripapillary hyperreflective ovoid mass-like structures (PHOMS). Macular volume scans with a pattern size of 20° X 20° and a total number of 25 B-scans were used to measure ganglion cell layer (GCL) volume. A peripapillary ring scan with a circle diameter of 12°, 3.4 mm, was used to evaluate retinal nerve fiber layer (RNFL) thickness. Automated perimetry (Octopus; Haag-Streit Diagnostics) was used for visual field examination, and medical health records (Epic) provided information regarding visual acuity, IOP, date of diagnosis, refraction, and vascular risk factors.

• **IMAGE ANALYSIS:** An EDI-OCT-scan was analyzed for each patient using Heidelberg Eye Explorer. Optic disc drusens were identified as hyporeflective structures with hyperreflective margins, positioned above the lamina cribrosa (LC), see Figure 1. The ODD were graded as “superficial” if predominantly located above the BMO and “deep” if predominantly located below the BMO. The ODD were graded as “large” if in a given eye, ODD > 200 µm in at least one direction were detected, and “small” if ODD measured < 200 µm in both directions. The mean BMO diameter was measured from 6 radial scans. In case of 24 radial scans every fourth scan was used to calculate mean BMO diameter. Prelaminar hyperreflective lines were noted. Peripapillary hyperreflective ovoid mass-like structures were identified as a peripapillary hyperreflective mass located at the rim of BMO.⁵ The PHOMS volume was measured using

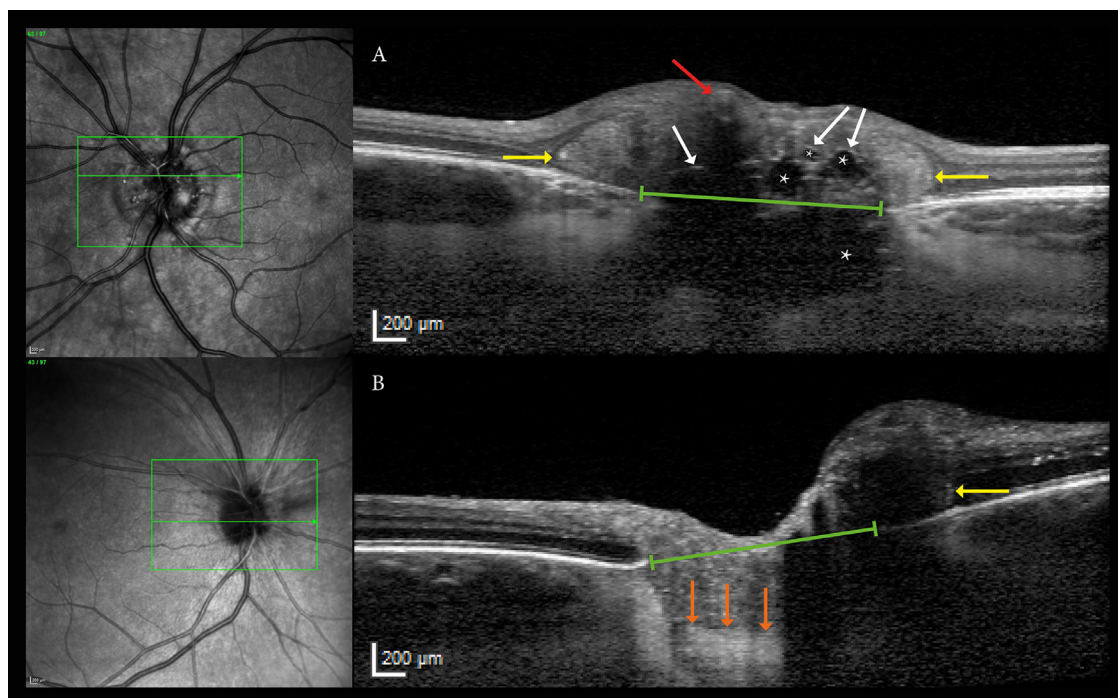


FIGURE 1. Enhanced depth imaging (EDI)-optical coherence tomography (OCT) of the optic nerve head (ONH). **A.** A 27-year-old female (left eye) with anterior ischemic optic neuropathy (AION) and optic disc drusen (ODD), termed ODD-AION. A fragment of a peripapillary hyperreflective ovoid mass-like structure (PHOMS, yellow arrows) is seen on both sides of Bruch's membrane opening (BMO). The ODD (asterisks) are hyporeflective structures with a hyperreflective margin (white arrows), and are seen both above (superficial) and below (deep) a green horizontal line connecting Bruch's membrane termination on either side of the optic disc; this line represents the BMO diameter. A blood vessel (red arrow) is also seen. **B.** A 56-year-old male (right eye) with AION and no ODD, termed nODD-AION. Segmental optic disc edema is seen, sparing the temporal side, and a nasal fragment of a PHOMS (yellow arrow), the BMO (green horizontal line), as well as the anterior border of lamina cribrosa (LC) (three orange arrows) are outlined.

a previously described method where total PHOMS volume = $2\pi^2abc$, (a = major radius of torus, b = horizontal cross-sectional radius, and c = vertical cross-sectional radius).⁹

The LC position was measured based on the criteria from Lee and associates¹⁰ in the central three-quarters of the optic disc using dense ONH EDI-OCT scans. Three perpendicular points from the anterior border of LC to the horizontal line from the BMO were drawn. The position of the perpendicular points was determined as follows: the longest distance between the anterior LC and BMO was set as the first line and two other lines were placed 100 and 200 μm away in a temporal direction, respectively. A mean based on the 3 points was calculated. The measurement was performed 6 times, if possible, for each eye and a final mean was used as the LC position. Four measurements were accepted as a minimum, otherwise that eye was excluded.

The GCL volume was read from the macular scan. The RNFL thickness was read via the circular scan, and thickness values for temporal superior (TS), temporal (T), temporal inferior (TI), nasal inferior (NI), nasal (N), nasal superior (NS), and the global value (G) were identified.

All images were analyzed by the same examiner (R.G.J) under guidance of expert examiners (S.H and L.L). The

PHOMS volumes were measured by an expert examiner (M.J).

• **STATISTICAL ANALYSIS:** Statistical analyses were performed using RStudio (version 1.41717, RStudio Inc). In cases of bilateral, sequential NA-AION within the inclusion period the first diagnosed eye was used as the study eye. In 2 cases of bilateral sequential NA-AION, 1 eye was diagnosed during the inclusion period, which is why 1 eye from each of these 2 patients was included for statistical analysis. In 6 cases of bilateral simultaneous NA-AION, the mean numerical and categorical values from the 2 eyes were used for statistical analysis. Test for normality was performed by visualizing the data using histograms and via the Shapiro test in RStudio. To compare normally distributed data, a 2-sample t test was used and data presented by means and standard deviations. In skew observations, a Wilcoxon (Mann-Whitney) test was used and data were presented by medians and interquartile range (IQR). Chi square test was used to compare categorical variables, except if any expected frequency was 5, then Fisher's exact test was used. The limit for statistical significance was set to $P < .05$.

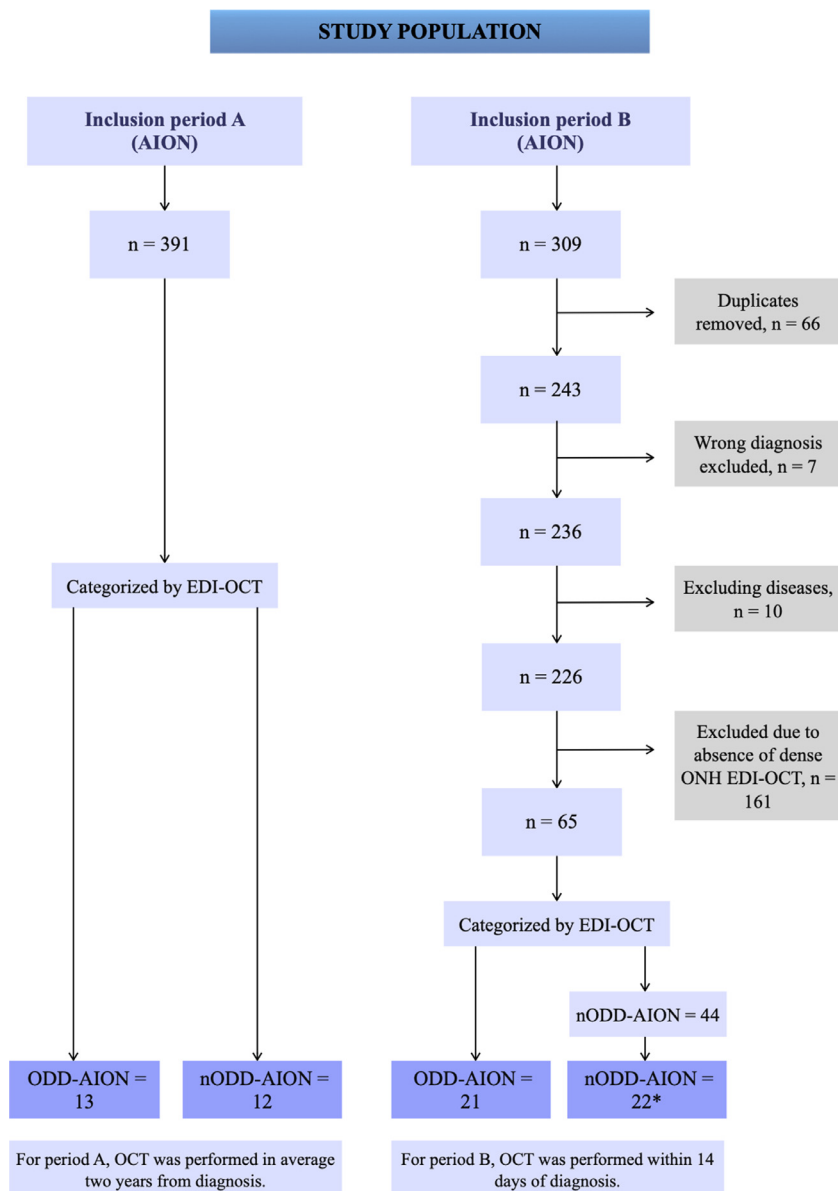


FIGURE 2. Flowchart demonstrating the categorization of study patients from inclusion periods A and B. AION = anterior ischemic optic neuropathy, EDI-OCT = enhanced depth imaging optical coherence tomography, nODD-AION = nonarteritic AION without ODD, ODD-AION = nonarteritic AION with optic disc drusen (ODD), ONH = optic nerve head.

* Of the remaining 44 with nODD-AION, the 22 with the most recent diagnosis were chosen in order to comment on a potential age difference at time of diagnosis between ODD-AION and nODD-AION patients.

The best corrected value was used for visual acuity. Visual acuity was measured with a Snellen chart in some patients; this was converted to ETDRS scores via the formula $85 + 50 \times \log(\text{Snellen fraction})$.

RESULTS

• **INCLUSION AND EXCLUSION OF PATIENTS:** For the selection process of patients in inclusion period A, see

Rueløkke and associates.³ Briefly, 391 patients with AION were identified. Of these, 14 eligible ODD-AION patients were identified and invited to participate, of which 11 accepted. During study examination, an additional 2 patients were identified as having ODD, though originally included as nOOD-AION patients, totaling 13 ODD-AION patients. Of the 391 AION patients, 377 nODD-AION patients were identified by coding and the 20 youngest, contactable, and eligible patients were invited. Three nODD-AION patients were identified in clinic. Eighteen of 23 invited patients wanted to participate, but 4 were excluded

TABLE 1. Demographic and Clinical Characteristics of Study Patients.

Demographic/Clinical Characteristics	ODD-AION (n = 34)	nODD-AION (n = 34)	P Value
Sex (%), male	25 (74)	24 (71)	1 $\exists$
Bilateral/unilateral, n	8/26	3/31	.1863 Ω
Age at diagnosis, yr	47.6 $\pm$ 16.81	56.9 $\pm$ 12.11	.0116 ϑ
Visual acuity, ETDRS letters	84 (77-86.5)	85 (75-88.25)	.7783 ψ
IOP, mmHg	14.0 $\pm$ 2.61	13.6 $\pm$ 3.33	.6578 ϑ
Automated perimetry MD, dB	-11.7 $\pm$ 7.15	-12.1 $\pm$ 5.81	.8239 ϑ
Refractive error, D	0 (-0.75 to 0.88)	0.3 (-0.19 to 0.94)	.4541 ψ

D = diopters, ETDRS = Early Treatment of Diabetic Retinopathy Study, IOP = intraocular pressure, MD = mean defect, nODD-AION = nonarteritic anterior ischemic optic neuropathy without optic disc drusen, ODD-AION = nonarteritic anterior ischemic optic neuropathy with optic disc drusen.

Ω = Fischer exact test, $\exists$ = Chi-square test, ϑ = 2-sample *t* test, ψ = Wilcoxon (Mann–Whitney) test.

Normally distributed data are represented as mean and $\pm$ standard deviation. Skewed observations are represented by median and interquartile range. Due to missing data, 10 patients were excluded in visual acuity analysis, 6 were excluded in IOP analysis, 6 were excluded in refraction analysis, and 1 was excluded from automated perimetry analysis.

Based on the following assumptions: if the date of diagnosis was not mentioned exactly but termed “spring,” the date for statistical analysis was set to March 1st. If only a year was listed for age at diagnosis, January 1st was used. A month has 30 days.

due to incomplete examination ($n = 1$), normal visual field test ($n = 1$), or detection of ODD ($n = 2$). Additionally, 2 patients were excluded from this study due to incomplete dense ONH EDI-OCT scans cf. ODDS scanning protocol. This process resulted in 12 patients with nODD-AION from inclusion period A.

From inclusion period B, 309 patients with AION diagnosed between November 30, 2017 and September 8, 2021 were identified via diagnostic coding; 66 duplicates were removed. Furthermore, 17 patients were excluded due to wrong diagnosis ($n = 7$), glaucoma ($n = 4$), arteritic AION ($n = 1$), ipsilateral unspecified ONH disease ($n = 3$), and pituitary adenomas ($n = 2$); 161 were excluded due to missing dense ONH EDI-OCT scan cf. ODDS-protocol. Of the 65 remaining AION patients from inclusion period B, 21 had ODD-AION. Of the 44 remaining patients with nODD-AION, the 22 subjects with the most recent diagnosis were selected.

In total, 34 ODD-AION and 34 nODD-AION ($n = 68$) were included, as shown in [Figure 2](#).

• **DEMOGRAPHIC AND CLINICAL CHARACTERISTICS:** Demographic and clinical characteristics of ODD-AION and nODD-AION patients are shown in [Table 1](#). There were no significant differences in sex, bilateral vs unilateral disease, visual acuity, IOP, perimetric mean deviation (MD), or refraction measurements between ODD-AION and nODD-AION patients. The ODD-AION patients were significantly younger than nODD-AION patients ($P = .0116$), with a mean age at diagnosis of 47.6 years for ODD-AION vs 56.9 years for nODD-AION patients.

• **OCT FINDINGS:** The OCT characteristics of ODD-AION and nODD-AION patients are shown in [Table 2](#). Twenty-eight of 34 (82%) ODD-AION patients had deep

ODD compared to superficial; 17 of 33 (52%) ODD were graded as large. It was found that ODD-AION patients had a significantly smaller BMO diameter (median 1419 μm) compared with nODD-AION patients (median 1596 μm) ($P = .031$). A significantly higher percentage of hyper-reflective lines was found in ODD-AION patients (94%) compared with nODD-AION patients (21%) ($P < .001$). A significantly higher percentage of PHOMS was found in ODD-AION patients (74%) compared with nODD-AION patients (30%) ($P = .00038$). However, the volume of PHOMS did not differ between the two groups ($P = .5721$). Finally, the position of LC was found to be deeper in patients with ODD-AION compared with nODD-AION; however this was not significant ($P = .1326$).

An overall macular GCL volume decrease was found in both ODD-AION and nODD-AION patients compared with what would be expected without an ischemic event, with no significant difference between the two groups ($P = .3315$), [Table 2](#). Likewise, an overall thinning of RNFL was found in both ODD-AION and nODD-AION patients compared with what would be expected without an ischemic event; the difference was not significant ($P = .4113$ for RNFL (G); global value), [Supplementary Table 1](#).

• **VISUAL FIELDS AND OCT FINDINGS: Comparisons Between the Two Inclusion Periods.** To investigate whether the difference in the timing of the autoperimetry and OCT scans in the 2 inclusion periods had any influence on the visual fields and OCT findings, a subanalysis was performed comparing data from the 2 inclusion periods. There was no significant difference in MD, ODD localization, BMO diameter, or GCL volume in ODD-AION patients from the 2 inclusion periods, [Supplementary Table 2](#). The RNFL was significantly thicker in inclusion period B compared with A,

TABLE 2. Optical Coherence Tomography Characteristics of Study Patients.

Optic Nerve Head Anatomy	ODD-AION (n = 34)	nODD-AION (n = 34)	P Value
Superficial/deep ODD	6/28	-	
Large/small ODD	17/16	-	
BMO diameter (IQR), μm	1419 (1336-1578)	1596 (1472-1696)	.0171 ψ
Hyperreflective lines, n (%)	31 (94)	7 (21)	< .001 $\exists$
PHOMS, n (%)	25 (74)	10 (30)	.0004 $\exists$
PHOMS volume, mm^3	0.3 $\pm$ 0.26	0.3 $\pm$ 0.16	.5721 ϑ
Lamina cribrosa position, μm	432.2 $\pm$ 95.79	389.8 $\pm$ 99.27	.1326 ϑ
GCL volume, mm^3	0.87 $\pm$ 0.19	0.81 $\pm$ 0.17	.3315 ϑ

AION = nonarteritic anterior ischemic optic neuropathy, BMO = Bruch's membrane opening, GCL = macular ganglion cell layer, nODD-AION = AION without ODD, ODD = optic disc drusen, ODD-AION = AION with ODD, PHOMS = peripapillary hyperreflective ovoid mass-like structures.

ψ = Wilcoxon (Mann-Whitney) test, $\exists$ = Chi-square-test, ϑ = 2-sample *t* test.

Normally distributed data are represented as mean and $\pm$ standard deviation. Skewed observations are represented by median and interquartile range. Only 22 ODD-AION patients and 24 nODD-AION patients were used for scleral canal diameter measurement, as the radial scan necessary for the measurement was not performed on the remainder. One was excluded from hyperreflective lines analysis due to non-identical categorical values in a bilateral case. One was excluded in the assessment of PHOMS presence due to opposing categorical values. In PHOMS volume analysis, 14 patients were excluded due to missing radial scan and 1 was excluded due to poor image quality. Eighteen patients were excluded in lamina cribrosa analysis due to either posterior ODD-border shadowing, optic disc edema shadowing, or poor OCT image quality. One was excluded in ODD size measurement due to opposing values in a bilateral case. Due to missing data, 33 patients were excluded in GCL volume analysis.

TABLE 3. Vascular and Ocular Risk Factors in Study Patients.

Vascular risk factors	ODD-AION (n = 34)	nODD-AION (n = 34)	P Value
Smoking, n (%)	6 (18)	9 (26)	.5586 $\exists$
Arterial hypertension, n (%)	5 (15)	15 (44)	.0166 $\exists$
Diabetes mellitus, n (%)	2 (6)	4 (12)	.6728 Ω
Hypercholesterolemia, n (%)	3 (9)	10 (29)	.0615 Ω
Sleep apnea, n (%)	1 (3)	1 (3)	1 Ω
Use of Sildenafil, n (%)	1 (3)	2 (6)	1 Ω
No risk factors, n (%)	22 (65)	11 (32)	.0153 $\exists$
Ocular risk factors			
Cataract surgery*, n (%)	6 (18)	2 (6)	.2585 Ω

AION = nonarteritic anterior ischemic optic neuropathy, nODD-AION = AION without ODD, ODD = optic disc drusen, ODD-AION = AION with ODD.

$\exists$ = Chi-square test, Ω = Fischer exact test.

Normally distributed data are represented as mean and $\pm$ standard deviation (sd). Skew observations are represented by median and inter-quartile-range (IQR).

*Cataract surgery before the onset of ODD-AION or nODD-AION.

as expected due to the more recent ischemic event in the former.

• **VASCULAR AND OCULAR RISK FACTORS:** A list of vascular and ocular risk factors is presented in Table 3. It was found that significantly more ODD-AION patients (65%) had no risk factors compared with nODD-AION patients (32%) ($P = .015$). A high prevalence of risk factors such as smoking, arterial hypertension, and hypercholesterolemia was found in nODD-AION patients. There was no sig-

nificant difference in the number of patients having had cataract surgery prior to NA-AION diagnosis between the 2 groups.

DISCUSSION

This study examined the OCT-based ONH anatomy of patients with ODD-AION compared with patients with

nODD-AION. This was performed to examine the hypothesis that anatomic ONH parameters aside from ODD characterize and potentially also play a role in the development of ODD-AION. It was found that ODD in ODD-AION were predominantly deep, but with an equal number of patients having large and small ODD. The ODD-AION patients presented more PHOMS and prelaminar hyperreflective lines and had smaller BMO diameters compared with patients with nODD-AION. Also, the ODD-AION patients were generally younger and had fewer vascular risk factors. These findings will be addressed and reflected upon in the following section.

The ODD-AION patients were significantly younger than nODD-AION patients, with a mean age at diagnosis of 47.6 years compared with 56.9 years. It has been proposed that ODD is an independent risk factor for NA-AION in patients aged < 50 years, as older patients more often suffer from RNFL loss and are thereby less likely to suffer from disc crowding.^{3,8,11,12} No differences in visual acuity, IOP, automated perimetry, or refraction were found between patients with ODD-AION or nODD-AION.

A very high number (82%) of patients with ODD-AION had deep ODD compared with superficial. In an age-matched (40-49 years) cohort of ODD patients without AION, 46 had deep ODD (54%) (data from⁹). Hence, deep ODD constitutes a risk factor for AION in patients with ODD. An explanation could be that the deep ODD increase the crowding of axons in the AION predilection site near the lamina cribrosa by themselves occupying space, exacerbating an incipient compartment syndrome. In theory, deep ODD could incite an ischemic event by compressing densely packed adjacent neurons and/or microvasculature, triggering inflammatory edema or infarction, and initiating the ischemic cascade. Also, patients with superficial ODD tend to have a thinner RNFL, and the axonal atrophy in these eyes could potentially have contributed to a decompression of the crowded disc-at-risk reducing the risk of AION. In ODD-AION patients, an equal number of patients had large (52%) and small (48%) ODD; this indicates that ODD size in itself is not a risk factor for ODD-AION. The overall conclusion that anatomical location, but not size, is important for the risk of developing acute ODD-associated visual dysfunction is in contrast to what has previously been found regarding the more slowly progressing ODD-associated visual field loss in ODD patients without NA-AION. Here size, but not anatomical location, is important and the larger the ODD, the more pronounced the visual field defect.¹³

This study found that ODD-AION patients had a smaller BMO diameter compared with nODD-AION patients. Both ODD and NA-AION have independently been linked to small cup-disc-ratios on fundus examination and small BMO diameters on OCT, but the literature is ambiguous.^{4,6,13} An early fundus photography-based analysis showed that patients with NA-AION had smaller scler-

al canals compared with healthy controls, with ODD patients presenting even smaller canals than NA-AION patients.¹⁴ Children with ODD have also been found to have smaller BMO diameters on OCT than healthy age-matched controls.¹⁵ This is in contrast to other studies showing no or even larger BMO diameters in ODD patients compared with controls.^{16,17} It has been argued that larger and superficial ODD can potentially cause a displacement of Bruch's membrane, thereby enlarging the scleral canal beyond its original width.^{15,16} The current study found an overrepresentation of deep ODD and no evidence of mechanical enlargement of Bruch's membrane in the ODD-AION patients. At this point, it cannot be concluded whether a small BMO diameter or the ODD themselves are the main risk factor behind ODD-AION, or if a combination of the 2 has any synergistic effect.

The ODDS Consortium has argued that prelaminar hyperreflective lines might be precursors for ODD in cases of *congenital* optic disc crowding.⁵ This was supported by a prospective study among children, where prelaminar hyperreflective lines were seen to progress into ODD at the 5-year follow-up examination.¹⁸ In the current study, 94% of the included ODD-AION patients had prelaminar hyperreflective lines, supporting the hypothesis of hyperreflective lines as precursors for ODD in case of congenital optic disc crowding. It has recently been shown that hyperreflective lines can remain as lines, without becoming ODD, in conditions with *acquired* disc crowding such as in papilledema where the crowding eventually clears.¹⁹ This indicates a pathological character of the hyperreflective lines. Interestingly, a rather large proportion (21%) of nODD-AION patients in the current study had prelaminar hyperreflective lines; this might have been due to acquired optic disc crowding as a consequence of the localized optic disc edema in parallel to what potentially occurs in the more generalized papilledema.¹⁷

A PHOMS is a peripapillary hyperreflective bulge of nerve fibers seen on OCT and usually caused by axoplasmic flow stasis.^{5,20,21} The ODDS Consortium has described PHOMS as a non-specific OCT sign seen in relation to either optic disc edema, ODD, or disc anomalies.²⁰ The presence and volume of PHOMS in ODD patients has been shown to decrease with age.⁹ The current study found a significant overrepresentation of PHOMS in the ODD-AION patients compared with nODD-AION patients. This finding is consistent with a recent study in which 15 of 28 (54%) ODD-AION patients had PHOMS compared to 8 of 29 (28%) nODD-AION patients.⁸ An explanation for this finding could be that the ODD-AION patients had PHOMS before the onset of AION and that the PHOMS remained after the occurrence of AION. However, it could also indicate that a PHOMS in itself could contribute to NA-AION, even though this is still highly speculative. The defining factor would be a prospective study following ODD patients before and after the onset of AION to closely study the development of PHOMS. The current study did not

find any significant difference in the volume of PHOMS between ODD-AION and nODD-AION patients. It is the clinical experience of the authors that the larger the optic disc edema, the larger the PHOMS. Therefore, the timing of the OCT scan in relation to the ischemic event could have an influence on the size of the PHOMS and thereby also potentially influence the axonal crowding leading to visual field defects. The PHOMS volume was not subanalyzed according to timing of OCT scans; however, no difference in visual field defects were detected when comparing ODD-AION patients from inclusion periods A and B, Supplementary Table 2.

While not significant, ODD-AION patients were found to have a deeper LC position compared with nODD-AION patients. A large portion of the eyes in this analysis of LC position had to be excluded due to posterior ODD shadowing in the ODD-AION group. Furthermore, the measurements were performed by a single examiner (RGJ); therefore, no correlation between interobserver vs intraobserver reproducibility was calculated. Despite these shortcomings, it is plausible that ODD cause remodelling and posterior displacement of LC to some extent, which could explain the difference in LC position between the two groups. These findings call for larger prospective studies to further evaluate the aspects of LC position in NA-AION pathogenesis.

The RNFL thickness and GCL volume are used as measures of axonal damage.⁶ The GCL has been shown to be more sensitive than RNFL when evaluating optic nerve damage.^{6,16} The current study found peripapillary RNFL thinning and macular GCL volume decrease in both groups, as expected after an ischemic event, but with no significant difference between the ODD-AION group and nODD-AION group.

This study found significantly more ODD-AION patients with no vascular risk factors compared with nODD-AION patients. Interestingly, the prevalence of vascular risk factors in the ODD-AION group was almost the same as the prevalence of vascular risk factors in the general population in Denmark. An overall prevalence of type 2 diabetes in Denmark of 4.4% has been measured,²² compared with 6% in the current ODD-AION patients. The prevalence of hypertension has been found to be 10.1% among people aged 40 to 49 years²³ compared with 15% in the current ODD-AION patients in the same age category. A prevalence of hypercholesteremia in Danish citizens aged ≥ 35 years without cardiovascular disease or diabetes of 32% has been measured.²⁴ The profile of hypercholesteremia seemed to match that of nODD-AION patients (29%), whereas the prevalence was lower in the current ODD-AION patients

(9%). These findings support the hypothesis that ODD is an independent risk factor for developing NA-AION in younger patients.

The current study found a tendency of ODD-AION patients to have had a higher incidence of cataract surgery prior to the development of NA-AION than nODD-AION patients; however, this was not significant. Cataract surgery as an ocular risk factor for NA-AION is not fully understood,^{25,26} yet an increased risk for post-cataract NA-AION has been seen in patients with fellow eye involvement.²⁵ The current study found that 2 of 6 in the ODD-AION group had fellow eye involvement prior to surgery. Further research is needed before it can be concluded whether ODD-AION patients suffer from more ONH-associated complications to cataract surgery than nODD-AION patients.

The main limitation of this study was that patients examined by Rueløkke and associates (inclusion period A) were all seen at the hospital after the onset of disease (an average of 2 years after diagnosis onset), yet the OCT images from the patients included via hospital records (inclusion period B) were from within 14 days of diagnosis during the acute stage, and thus more likely to still have optic disc edema, and thereby elevated RNFL, as opposed to RNFL thinning, which is seen when the optic disc edema has subsided revealing the resulting atrophy. It could be argued that the lack of personalized keratometry in this study could have induced a possibility of OCT measurement error, thereby constituting a limitation. However, as there was no difference in refraction between ODD-AION and nODD-AION patients, significant measurement errors due to transverse magnification were highly unlikely. No adjustment for transverse magnification in OCT measurements was therefore performed.

It is believed that this is the first study to analyze and compare anatomical and vascular risk factors in ODD-AION and nODD-AION patients. While size, but not anatomical location, is important for the gradual development of visual field defects in patients with ODD,¹³ it is the opposite case for ODD-AION, where deep ODD is a risk factor but size is unimportant. This study has shown that ODD-AION patients are younger at the time of diagnosis and have fewer vascular risk factors compared with nODD-AION patients, which is consistent with existing literature, suggesting that ODD is an independent risk factor for developing NA-AION. This establishes the importance of subdividing patients in ODD-AION and nODD-AION when searching for NA-AION risk factors. Finally, the results of this study suggest PHOMS as a possible contributing anatomical risk factor for the development or exacerbation of NA-AION.

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